AARUPADAI VEEDU INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

INTERNAL ASSESSMENT - 2024-2025

Degree & Branch:	OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES	**Regulations:**	2021 (Autonomous)
Course Code & Title:	35021P28 - R Programming	**Semester / Year:**	IV Semester / 2024-2025
Date & Session:	FN Session	**Duration & Marks:**	90 Minutes | Max. 50 M

Instructions: Answer all questions according to section regulations.

PART - A

Q.No	Question Text	CO	Bloom's	Marks
1	Explain the features of R and describe the steps involved in creating, executing and saving an R script with suitable examples.	CO1	K2 - Understand	2
2	Explain the characteristics of lists in R and discuss how they differ from atomic vectors and matrices. Describe the operations used to create, index, and apply functions to recursive lists with code examples.	CO2	K4 - Analyze	2
3	Explain the internal representation and utility of factors in R. Describe how frequency tables and cross-tabulations are constructed and analyzed using `table()`, `prop.table()`, and `margin.table()` with practical code demonstrations.	CO3	K4 - Analyze	2
4	Compare the S3 and S4 object systems in R using appropriate architectural criteria. Discuss their differences in formal class definition, type validation, method dispatch, and slot encapsulation.	CO4	K4 - Analyze	2
5	Explain the theoretical assumptions underlying classical Ordinary Least Squares (OLS) linear regression. Describe how each assumption is diagnosed in R using residual plots, the Shapiro-Wilk test, and the Durbin-Watson statistic.	CO5	K4 - Analyze	2

PART - B

Q.No	Question Text	CO	Bloom's	Marks
6	Distinguish between vectorized operations and scalar iterations using loops in R. Illustrate your answer with execution examples comparing their syntax and computational performance.	CO1	K4 - Analyze	5
7	Explain the characteristics of lists in R and discuss how they differ from atomic vectors and matrices. Describe the operations used to create, index, and apply functions to recursive lists with code examples.	CO2	K4 - Analyze	5
8	Explain the internal representation and utility of factors in R. Describe how frequency tables and cross-tabulations are constructed and analyzed using `table()`, `prop.table()`, and `margin.table()` with practical code demonstrations.	CO3	K4 - Analyze	5
9	Compare the S3 and S4 object systems in R using appropriate architectural criteria. Discuss their differences in formal class definition, type validation, method dispatch, and slot encapsulation.	CO4	K4 - Analyze	5

PART - C

Q.No	Question Text	CO	Bloom's	Marks
10(a)	Part I (Programming - 6 Marks): Write a modular, fully commented R script/function to implement an end-to-end workflow for Introduction to R. Your code must handle vector inputs, validate data structures, and display the transformed results. Part II (Descriptive - 4 Marks): Analyze the computational efficiency, vectorization mechanisms, and asymptotic memory trade-offs of your implementation compared to iterative loops in base R.	CO1	K3	10

10(b)	Part I (Programming - 6 Marks): Write a modular, fully commented R script/function to implement an end-to-end workflow for Matrix operations. Your code must handle vector inputs, validate data structures, and display the transformed results. Part II (Descriptive - 4 Marks): Analyze the computational efficiency, vectorization mechanisms, and asymptotic memory trade-offs of your implementation compared to iterative loops in base R.	CO2	K3	10
11(a)	Part I (Programming - 6 Marks): Write a modular, fully commented R script/function to implement an end-to-end workflow for Merging Data Frames. Your code must handle vector inputs, validate data structures, and display the transformed results. Part II (Descriptive - 4 Marks): Analyze the computational efficiency, vectorization mechanisms, and asymptotic memory trade-offs of your implementation compared to iterative loops in base R.	CO3	K4	10
11(b)	Part I (Programming - 6 Marks): Write a modular, fully commented R script/function to implement an end-to-end workflow for Input/Output. Your code must handle vector inputs, validate data structures, and display the transformed results. Part II (Descriptive - 4 Marks): Analyze the computational efficiency, vectorization mechanisms, and asymptotic memory trade-offs of your implementation compared to iterative loops in base R.	CO4	K4	10

Course Outcome (CO) & Cognitive Level Distribution Summary

CO	Total Marks	Weightage (%)
CO1	17 M	34.0%
CO2	7 M	14.0%
CO3	17 M	34.0%
CO4	7 M	14.0%
CO5	2 M	4.0%